Causal Quantum Gravity: A Graph-Laplacian Spine for Unified Quantum–Relativistic Dynamics and a Discrete Causal Signature

Yaoharee Lahtee

Open Civil Science Initiative

July 5, 2026

Abstract

We report a tier-honest, machine-checked (Coq) result within the *readout-not-truth* discrete-substrate framework: a single graph-Laplacian "mother equation" $M\partial^2\Phi + D\partial\Phi + KL_R\Phi + \nabla V(\Phi) = J - \eta$ yields, without importing external physics formulas, (i) a quantum energy-dispersion relation identical in algebraic form to the relativistic (Klein–Gordon-family) mass-shell condition, and (ii) a discrete causal/Lorentzian bilinear form whose continuum limit is the d'Alembertian $\Box$. We prove, for the first time in this program, that these two readouts are not independent: the quantum dispersion condition is *exactly* (pure $\mathbb{Q}$-algebra, no continuum limit) the vanishing condition of the same boost-invariant wave operator that governs the relativistic branch — one equation, two readouts. We further report an exhaustive, adversarially-audited finding that general relativity (continuum curvature, the Schwarzschild solution, Einstein's field equations) is *not* derivable from this root, and argue — consistent with the framework's own injected-infinity diagnostic — that this is the philosophically correct outcome, not a gap. In its place we exhibit a genuinely native, axiom-free discrete curvature object (Forman–Ricci curvature on the graph) and prove its exact algebraic link to an independently-proven stability theorem. A companion finite-domain Perfectly-Matched-Layer numerical bridge reproduces a literature quasinormal-mode frequency to ∼0.1–1.2%, honestly tiered as `finite_diagnostic`, not `Th_coqc`. Responding to an independent adversarial referee review by proving more rather than retreating any claim, we further report six results promoting previously informal stances to theorems: an exact frequency/UV ceiling forced by the graph's own maximum degree, an exact "no local creation" energy-balance law, a Schrödinger-shaped skew-adjoint first-order skeleton, a causal sign-construction theorem (with an honestly disclosed partial closure), and a discrete Noether theorem; and one further result showing that growing the graph itself is a native, non-continuum discrete analog of cosmological expansion — curving space negatively, conserving energy exactly, and diluting spectral modes by a closed-form law, while explicitly not claiming to resolve any general-relativistic cosmological discrepancy. A further six files add an action-stationarity identity, a curvature-energy-horizon structure, and a four-step spectral-convergence ladder resolving the flat/diagonal-metric case of the mother equation's continuum limit in arbitrary dimension. A further eleven files add a discrete disk-before-lock analog, close a seed-to-spectral-ceiling chain, bridge to an algebraic track (curvature-Noether invariance, a rotation-matrix identity, a golden-ratio-adjacent pentagon spectrum), a graph-native boundary-energy fact and a degree-curvature bound, a general tensor-reconstruction fact with a curvature bridge conditional on one stated hypothesis, and a momentum- optimizer/kernel-energy identification. We further compose several of these theorems — none individually about mass — into a curvature-limited mass bound conditional on a single disclosed, non-theorem identification, whose contrapositive states that sufficiently large mass forces negative combinatorial curvature; a companion synthesis note included as a supplement gives the full derivation and literature positioning. Two further files restate the degree/frequency-ceiling family in entropy-flavored language (explicitly disclosed as a naming choice, not a physical claim) and prove a graph-native boundary-screening theorem (information confined to a region's strict interior cannot reach a fixed exterior weighting), each explicitly scoped against holographic or general-relativistic over-reading. All told: 40 Coq files, 186 theorems, every one machine-checked. Every claim below is accompanied by its tier, its exact source theorem, and a reproduction command.

Keywords: discrete quantum gravity; graph Laplacian; retained-readout framework; Klein–Gordon unification; Forman–Ricci curvature; discrete Noether theorem; graph growth; Coq formal verification; quasinormal modes; reproducibility.

Preprint notice. This manuscript is a preprint (v1.0) and has not undergone formal peer review. It has, however, undergone one internal editorial pass and one independent adversarial technical review (§23), both logged. Definitions, proofs, and numerical results are reproducible from the accompanying repository and remain provisional until independently reproduced by a third party.

1 Introduction

This manuscript reports work within a larger research program (the "readout-not-truth" framework) whose central commitment is that a formal system should never assert more than it has machine-checked: a claim is `Th_coqc` only if `coqc` accepts it and `Print Assumptions` reports the global context closed over $\mathbb{Q}$; `+reals` if it additionally depends on Coq's disclosed Reals axioms; `finite_diagnostic` if it is a reproducible numerical measurement, not a proof; `Dr` if it is an interpretive stance; `Open` if it is an admitted gap. Collapsing these tiers — describing a measurement as a proof, or a stance as a derivation — is treated as the framework's cardinal error.

The object under study is a single *mother equation* on a finite graph (Eq. 1), built from one primitive, δ_R (retained difference), via a graph Laplacian L_R. This paper asks, and answers honestly: what, exactly, follows from this one equation, and what must be imported?

2 Claim Tiers

Tier	Meaning
`Th_coqc`	Machine-checked, axiom-free over $\mathbb{Q}$; `Print Assumptions` reports the context closed.
`+reals`	`Th_coqc` but depends on Coq's disclosed Reals axioms (`sig_forall_dec`, `functional_extensionality_dep`).
`finite_diagnostic`	A reproducible numerical measurement; not a proof.
`Dr`	An interpretive stance, not machine-checked.
`Open`	An admitted, undischarged gap.

Table 1: Claim tiers used throughout this manuscript.

3 The Mother Equation

$$M\partial^2\Phi + D\partial\Phi + KL_R\Phi + \nabla V(\Phi) = J - \eta \tag{1}$$

where L_R is a weighted graph Laplacian built from edge data (u_e, v_e, w_e), M, D, K are inertia/dissipation/coupling scalars, and $J - \eta$ is a drive-minus-loss source term. All results below concern the *conservative* sector ($D = 0$, $\nabla V = 0$, $J = \eta$) unless stated otherwise.

4 Branch 1: Quantum Readout

Source. `formal/InfoSchrodinger.v` (extracted from the audited core; `Th_coqc`). A temporal mode gives $\partial^2 \to -\omega^2$; on an L_R-eigenmode of eigenvalue λ the spine residual is

$$K\lambda - M\omega^2 = 0 \iff M\omega^2 = K\lambda. \tag{2}$$

Composing with the Planck–Einstein relation $E = \hbar\omega$ [1, 2] gives

$$E^2 M = \hbar^2 K\lambda. \tag{3}$$

Validation (`finite_diagnostic`). Feeding the C_6 (benzene-ring) graph Laplacian spectrum through Eq. 3 reproduces the qualitative structure of Hückel molecular-orbital theory [5] (adjacency spectrum $\{2, 1, 1, -1, -1, -2\}$, degeneracy pattern 1-2-2-1); the resonance energy 2β is graph-theoretic, its numeric value in eV requires the empirical β. This is a consistency demonstration, not an independent confirmation of a fitted constant.

5 Branch 2: Special-Relativistic Readout

Source. `formal/InfoLorentz.v`, `InfoLorentzContinuum.v` (`Th_coqc` Tier-0 and `+reals` respectively; both independently re-verified via `Print Assumptions` in this work, though the underlying proofs were authored 2026-06-27, prior to this manuscript's other results). A signed bilinear form on graph edges,

$$\mathrm{cform}_{\mathrm{sgn}}(x, y) = \sum_e \mathrm{sgn}(e)\, w_e\, \delta_x(e)\, \delta_y(e), \tag{4}$$

is self-adjoint, reduces to L_R's own quadratic form when $\mathrm{sgn} \equiv +1$ (Eq. 5), and is invariant under permutation of the edge list (Eq. 6):

$$\mathrm{cform}_{+1}(x, y) = \mathrm{info\_form}(x, y), \tag{5}$$

$$\mathrm{cform}_{\mathrm{sgn}}(x, y; \pi(E)) = \mathrm{cform}_{\mathrm{sgn}}(x, y; E). \tag{6}$$

The continuum limit of the corresponding signed second-difference operator is proved (`+reals`) to be the d'Alembertian

$$\Box = -\partial_{tt} + \partial_{xx}. \tag{7}$$

Scope. The sign function sgn in Eq. 4 is universally quantified in the source theorems, not itself derived from a partial causal order on the graph nodes; a genuine derivation of an indefinite signature from δ_R's own (currently linear) order relation is an open research direction (§27), not claimed here. Separately, the specific Lorentz boost formula $\gamma(t - vx)$ [4, 3] is an imported `Definition`, verified (not derived) to leave $\Box$ invariant.

6 Unification: Branches 1 and 2 Are One Equation

This section reports the manuscript's principal new result. Define, following the already-proven boost-invariant exact quadratic wave operator box_quad$(a_{tt}, a_{xx}) := -2a_{tt} + 2a_{xx}$, and the spine residual $R(M, K, \omega^2, \lambda) := K\lambda - M\omega^2$ from Eq. 2. Under the exact reparametrization $a_{tt} := M\omega^2/2$, $a_{xx} := K\lambda/2$:

$$\text{box\_quad}\Big(\tfrac{M\omega^2}{2}, \tfrac{K\lambda}{2}\Big) = R(M, K, \omega^2, \lambda). \tag{8}$$

This is a pure ring identity (`ring` tactic, zero-line proof); it is not a coincidence of notation but a statement that both objects were built from the same signed second-difference structure. Consequently,

$$M\omega^2 = K\lambda \iff \text{box\_quad}\Big(\tfrac{M\omega^2}{2}, \tfrac{K\lambda}{2}\Big) = 0, \tag{9}$$

i.e. **the quantum dispersion condition (Branch 1) is exactly the vanishing of the same wave operator (Branch 2)**. Because box_quad is already proven boost-invariant, this condition is preserved under the identical Lorentz boost:

$$\text{box\_quad}\big(c_{tt}(g, v, \ldots),\, c_{xx}(g, v, \ldots)\big) = 0 \tag{10}$$

whenever Eq. 2 holds and $g^2(1 - v^2) = 1$. All three equations are `Th_coqc` (`formal/InfoQuantumRelativityUnification.v`).

Interpretation. The dispersion relation $E^2M = \hbar^2K\lambda$ (Eq. 3) is quadratic in E, structurally matching the relativistic mass-shell relation $E^2 = (pc)^2 + (mc^2)^2$, not the linear-in-E non-relativistic Schrödinger dispersion. Equations 8–10 show this is not an incidental resemblance: it is because the mother equation's own second-order-in-time structure is, algebraically, the same object as the relativistic wave operator whose boost invariance Branch 2 already establishes. Branches 1 and 2 are therefore one readout of one equation, not two separately-argued branches sharing a name.

7 Branch 3: General Relativity — Honestly Not Derived

An exhaustive audit of every gravity-touching module in the source repository found no path from the mother equation to continuum general relativity; every module either imports the Schwarzschild metric factor, Einstein's field equations, the Unruh temperature, or the Bekenstein–Hawking entropy as an external `Definition` (self-disclosed in each module's own header), or formalizes generic tensor algebra over free variables never connected to L_R.

Eight independent attempts to close this gap were tested this program and refuted or left open: a bare definitional alias; a dimensionally-forced numerological coincidence (shown to equally "confirm" an unrelated quantity, hence non-discriminating); an axiom-restatement; a structurally mass-independent prediction contradicted by a ten-billion-fold mass comparison against the true $\kappa \propto 1/M$ scaling; a partial WKB match; two hyperboloidal-compactification attempts that hit a genuine *irregular* singular point at spatial infinity (an essential singularity, re-injecting exactly the actual-infinity artifact this framework's own diagnostic refuses); and one success (§9) that shares only a discretization *method*, not an equation, with the mother equation.

Conclusion. Continuum general relativity — curvature defined via derivative limits on a complete manifold — is, by this framework's own injected-infinity diagnostic, a non-readout. Its absence from this derivation is therefore the philosophically correct outcome, matching the unresolved state of every other discrete-substrate research program surveyed (causal sets [12], the Wolfram Physics Project [13], Regge calculus [14]).

8 Branch 4: Native Discrete Curvature

Source. `formal/InfoDiscreteGraphCurvature.v` (`Th_coqc`, axiom-free). Rather than chasing continuum curvature, we exhibit a discrete curvature notion — Forman–Ricci curvature [6] — that requires no derivative, limit, or manifold, and is already a native function of L_R's own edge/degree data:

$$F(u, v) = 4 - \deg(u) - \deg(v). \tag{11}$$

We prove that under uniform edge weight w, the weighted degree wdeg (from an independently-proven coercivity/stability theorem) equals $w \cdot \deg$ exactly:

$$\text{wdeg}(i) = w \cdot \deg(i) \implies C_{\text{safe}}V_{\max}\,\text{wdeg}(i) = C_{\text{safe}}V_{\max}w\,\deg(i). \tag{12}$$

The same degree count that sets an edge's curvature (Eq. 11) also sets, by exact substitution, the dissipation a node requires for the spine to remain coercive. We note, following adversarial review, that this link is exact but shallow: any monotone function of degree would exhibit the same shared dependence (§23).

9 Bridge: Quasinormal-Mode Numerics

Source. `formal/InfoQuantumGravityRootBridge.v` (`+reals`) + `scripts/verify_quantum_gravity_root_bridge.py` (`finite_diagnostic`). The Regge–Wheeler equation [8]

$$\frac{d^2\psi}{dr_*^2} + [\omega^2 - V(r)]\psi = 0 \tag{13}$$

built on the already-verified Schwarzschild metric factor $f(r) = 1 - 2M/r$, is discretized as a finite path-graph Laplacian eigenvalue problem with a Perfectly-Matched-Layer absorbing boundary [11] — no point at infinity anywhere. This converges ($N = 1600$–6400) to

$$M\omega \approx 0.4841 - 0.0956\,i, \tag{14}$$

matching the literature scalar $\ell = 2$, $n = 0$ fundamental quasinormal-mode value $M\omega \approx 0.4836 - 0.0968\,i$ [9, 10] to ∼0.1% (real part) / ∼1.2% (imaginary part), robust across grid resolution, absorbing-layer strength, and domain size.

10 A Strengthening Campaign: Six Results Closing Referee-Flagged Gaps

Following the same referee review, six further results were added, each promoting a previously informal (`Dr`-tier) stance to a `Th_coqc` theorem — by proving more, not retreating any existing claim. Every file: pre-verified by exact-rational numerical testing before authoring, then independently compiled and `Print Assumptions`-checked here; all Tier-0 axiom-free; zero `funext`, zero classical axioms, zero `admit`.

Frequency ceiling and CFL window (`InfoSpectralCeiling.v`, `InfoRecurrenceEnergy.v`, `InfoQuantumFrequencyCeiling.v`, 20 theorems). A pure degree-sum argument — no eigenvalue theory, no square root — gives $\lambda \leq 2d_{\max}$, a Rayleigh-quotient form of the Gershgorin bound. Composed with Branch 1's dispersion relation, this gives a hard frequency ceiling

$$M\omega^2 \leq K \cdot (2d_{\max}), \tag{15}$$

forced by the graph's own maximum degree, not asserted as a physical constant. Composed further with an exact discrete-leapfrog Lyapunov identity, the same degree bound guarantees both a CFL-style stability window $0 \leq a \leq 4$ and energy-non- increasing dynamics under dissipation. This replaces an earlier interpretive "τ_c floor" stance with an exact structural theorem about what sets it.

No local creation (`InfoGraphFluxBalance.v`, 8 theorems). A discrete divergence theorem plus Green's identity (summation-by-parts) prove that L_R's coupling term contributes exactly zero to the mother equation's total energy budget — it telescopes to zero across every edge. Energy can only change via dissipation (≤ 0) or source work.

A Schrödinger-shaped skeleton (`InfoCompanionSkew.v`, 5 theorems). Writing the conservative sector as a first-order companion system $\Psi = (x, v)$ with generator $B(x, v) = (Mv, -KLx)$, the generator is proved skew-adjoint under the energy inner product $\langle(x, v), (y, w)\rangle_E = K \cdot \mathrm{gform}(x, y) + M\langle v, w\rangle$, moving every state exactly orthogonal to its own energy level set. This is the algebraic core of norm-preserving first-order evolution — the $i\hbar\partial_t\psi = H\psi$ skeleton without i, without $\mathbb{C}$, without $\sqrt{\cdot}$. (One of the five, `green_identity`, is a helper lemma re-proved inside this file so it checks standalone, not a fifth headline result beyond the four above.) This "without i" skew-adjoint skeleton is a separate, weaker claim than §16's later, narrower observation that the mother equation's own stepper matrix literally IS the standard i-rotation matrix at one specific stability-window parameter value: that later claim never invokes $\mathbb{C}$ or Hilbert-space structure either, and is scoped to a single real 2×2 matrix identity, not to the Schrödinger-shaped skeleton above. The two do not conflict, but should not be read as reinforcing each other without this note.

Causal signature (`InfoCausalSignature.v`, 7 theorems). §5 disclosed that sgn is a free parameter. This file proves a sign function can instead be *constructed* from a relation's comparability (comparable pairs → −1, incomparable pairs → +1), yielding an exact PSD-minus-PSD split, two lightcone- style cone inequalities, and, on a concrete star-graph example, a genuine rational- congruence witness of Minkowski type $(1, 3)$. **This does not fully close the gap.** This repository's own causal order (built from a strict total order on its domain, proved order-isomorphic to $\mathbb{N}$) admits no incomparable pairs at all — applying the construction to it would degenerate to an all-timelike signature, not the indefinite structure demonstrated on the constructed example. The construction is real and general; it does not yet connect to this repository's own specific causal order, which would require a genuinely richer (non-total) structure.

Discrete Noether (`InfoGraphNoether.v`, 7 theorems). Given a graph automorphism σ, the Laplacian commutes with it, every quadratic structure built from L_R is σ-invariant, and the antisymmetric pairing $W(p, q) = \sum_i p(\sigma i)q_i - p_i q(\sigma i)$ is exactly conserved under the conservative step: genuine Noether shape, the inertial term cancels pointwise and the coupling term dies by equivariance plus summation-by-parts. A concrete instance (the 6-cycle with a rotation automorphism) confirms the theorem is non-vacuous.

11 Graph Growth: A Native Discrete Expansion Analog

A natural question, raised independently, is whether this framework can address a real cosmological discrepancy such as the Hubble tension (the $\sim$ 5–6σ disagreement between early-universe and local-distance-ladder measurements of H_0). The honest answer is no: that is a general-relativistic/FRW-cosmology problem, and §7 establishes this repository derives no general-relativity content at all. Forcing a connection would repeat the same category error as the eight refuted GR attempts of §7.

Following the same discipline that produced Branch 4 (curvature, §8 — a native discrete object in place of an unreachable continuum target), a natural discrete analog of *expansion* was investigated instead: growing the graph itself. `InfoGraphGrowth.v` (13 theorems, axiom-free) proves, for the mother equation's own graph structure:

Growth curves space negatively. Adding an edge e^* to a graph strictly decreases the Forman curvature (§8) of every edge adjacent to e^* by at least 1, exactly:

$$F_{e^*::E}(e) = F_E(e) - \mathrm{share}(e^*, u) - \mathrm{share}(e^*, v). \quad (16)$$

"Space creation curves space negatively" is a combinatorial identity, not an analogy.

Growth has an exact energy account. Adding an edge changes the vector energy balance (§10) by exactly h^2K times the product of the field difference across the new edge; a field continuous across the new edge changes the energy not at all (adiabatic growth).

The order-collapse obstruction, proved. §10's causal- signature gap — this repository's causal order is total, so it cannot produce an indefinite signature — is here promoted from an observation to a theorem: for *any* graph where every edge is comparable under the order, the signed form collapses to $-\mathrm{gdiag} \leq 0$ everywhere. "Space is incomparability; a total order has no space" is now exact, not descriptive. The qualitative statement — that a total causal order admits no spatial structure — is established folklore in causal set theory [17]; a literature check found no prior published proof of this specific curvature-collapse form, but the underlying intuition is not new to this work and is cited as such.

An exact dilution law. On a ramp/path growth sequence, the Rayleigh quotient of a specific test state has closed form $12/(n(n + 1))$ — division-free throughout, via an $\mathbb{N} \to \mathbb{Q}$ power-sum injection. Paired with the frequency ceiling (Eq. 15), the growing graph brackets its own spectrum from both sides: a ceiling set by degree, a floor that dilutes as $\sim 1/n^2$.

Scope (`finite_diagnostic`, honestly bounded). No exponential or de Sitter-type dilution law was found; only power laws were observed, and a star-growth sequence does not dilute at all — the dilution law is a property of the growth *topology*, not of growth per se. This is a native, non-continuum, discrete expansion analog: growth simultaneously curves space negatively, has exact energy accounting, and dilutes spectral modes with a closed-form law. It is not a cosmological expansion rate, does not reproduce H_0 under any interpretation, and Hubble-parameter matching remains exactly the non-readout target this repository does not chase.

12 Action Stationarity and a Curvature-Energy-Horizon Structure

Two further results (`InfoActionStationarity.v`, 8 theorems, and `InfoCurvatureBalance.v`, 12 theorems, both Tier-0 axiom-free) are reported here plainly, for exactly what they prove — neither inflated nor understated.

The mother equation is an exact stationarity readout. Define the discrete window action $S = \sum_n \left[\frac{M}{2}\left(\frac{\Delta x_n}{h}\right)^2 - \frac{K}{2}\mathrm{gform}(x_n, x_n)\right]$. Perturbing a single node by s changes the action by an exact quadratic identity (`action_polarization`); on the conservative step, the linear coefficient of that perturbation — the discrete analogue of $\partial S/\partial x_n$, computed via polarization in $\mathbb{Q}$, with no limit — vanishes identically at every node (`mother_step_stationary`). The mother equation *is* the discrete Euler–Lagrange stationarity condition of this action, exactly, not by analogy. Two further facts sharpen this: `damped_step_balance` shows dissipation is *exactly* the obstruction to stationarity (the first variation under the damped step is $2sc(q_i - p_i)$, vanishing only when $c = 0$); `strict_descent` shows that whenever the step residual is nonzero, an explicit variation ($s = R/C$) strictly decreases the action — non-solutions are never selected, a genuine, exact argmin content, not a metaphor.

What this does and does not establish. This is a real, native fact about the mother equation, worth stating in full: an argmin/stationarity reading of the dynamics that was previously informal is now an exact theorem. It is reported here at full strength. What it does *not* do is narrow any gap toward general relativity specifically. Being expressible as the stationarity of an action is a property shared by essentially every Lagrangian physical theory — classical mechanics, Maxwell's equations, and general relativity itself all share it. Two theories both admitting a variational formulation is not evidence that one derives the other; it is the most generic structural property a physical law can have. This result stands on its own merits and is reported at that value, no more and no less. A literature check (2026-07-05) found this file's specific combination — one theorem for the matter/field side on a fixed graph, a separate theorem for an edge-addition cost-benefit law on the graph's own topology — is not identical to any single prior construction, but is closely related to Trugenberger's Combinatorial Quantum Gravity [16], which varies a statistical action over dynamical random graphs where matter arises as curvature excitations and geometry emerges from graph condensation, with the Einstein–Hilbert action recovered in a geometric phase. The relation is noted explicitly rather than left for a reader to discover: this file's two separately-proved stationarity statements are a different formalization choice from Trugenberger's single combined action, not an independent invention of the underlying idea of varying both matter and graph topology together.

A curvature-energy-horizon structure. `InfoDiscreteGraphCurvature.v`'s degree identity composes exactly with the growth results of §11: $\deg(u) + \deg(v) = 4 - F(u, v)$ (`edge_deg_curvature`), so the coercivity dissipation threshold from §8 is an exact affine function of Forman curvature (`horizon_threshold_curvature`) — more negative curvature forces a strictly higher threshold (`threshold_antitone`). Taking the per-edge retention benefit to be affine in curvature, $b(e) = \alpha + \beta F(e)$ — an explicit modeling choice, not independently derived from anything else in this framework — gives an exact two-way iff between edge-retention and curvature (`curvature_balance_add`, `curvature_balance_rem`): under that choice, energy strain and curvature determine each other exactly. Separately, and without depending on that ansatz, a genuine dichotomy is proved directly from the dissipation-threshold structure: `no_escape`/`no_escape_strict` show that once per-node dissipation exceeds every admissible control's budget, no choice can prevent energy from being non-increasing; `repair_exists` shows the converse — below that threshold, a control exists that keeps a local mode from decaying. This gives a principled (non-circular) answer to an earlier open question about how to fix the per-node horizon threshold: it is exactly the point at which dissipation exceeds every admissible selection's budget, and this budget is set by curvature. `vacuum_joint_stationary` confirms non-vacuity: a constant field is a joint stationary point of both the field and (under the curvature-affine ansatz) the geometry side, on every graph.

Scope, stated plainly. The horizon dichotomy (`no_escape`/ `repair_exists`) is unconditional and exact. The curvature-balance iff (`curvature_balance_add/rem`) is exact conditional on the affine-benefit ansatz — a real theorem about a chosen model, not a derivation of why that model is the correct one. Neither result derives Einstein's field equations, the specific coupling constant, or dynamical geometry sourced by matter in general; neither is claimed to.

13 The n-Dimensional Spectral Convergence Ladder

Whether L_R spectrally converges to a continuum Laplace–Beltrami operator in more than one dimension has been an open frontier since Branch 4 (§8). Four further files (`InfoProductSpectrum.v`, 7 theorems, Tier-0; `InfoContinuumLimit_nD.v`, 4 theorems, Tier-1; `InfoWeightedReadout.v`, 2 theorems, Tier-1; `InfoCrossTermDominance.v`, 9 theorems, Tier-0) factor this into four steps rather than attempting it as one result, each tagged at its true tier.

Step 1 (Tier-0): Kronecker/Rayleigh additivity. An n-dimensional grid is the Cartesian product of 1-D chains, and a product graph's quadratic form separates exactly: $\mathrm{pgform}(x \otimes y) = \mathrm{gform}_G(x)\|y\|^2 + \|x\|^2\mathrm{gform}_H(y)$ (`form_separates`), with the analogous exact identity for the norm (`norm_separates`). Consequently a Rayleigh pair on each factor lifts to a Rayleigh pair on the product with eigenvalues adding (`rayleigh_additive`) — pure $\mathbb{Q}$ combinatorics, no eigenvalue theory or continuum limit invoked.

Step 2 (Tier-1, `sig_forall_dec`+`funext` only, confirmed no `classic` leak): the flat multi-axis readout. Reusing this repository's own 1-D symmetric-second-difference machinery (`tends0`, verbatim from the audited core) with two new closure lemmas (additivity, scaling), `weighted_two_axis_readout` shows a weighted sum of per-axis second-difference quotients converges to the same weighted sum of the per-axis limits. `box_2d_readout` (weights $-1, 1$) reproduces this repository's own `lorentz_box` construction as an instance — confirming the extension is consistent with the existing kernel, not a divergent reformulation. `laplacian_2d_readout`/`laplacian_3d_readout` give the flat n-D Laplacian.

Step 3 (Tier-1, same axiom profile): the variable-coefficient case. The flux-form stencil $a(x{+}h/2)(f(x{+}h){-}f(x)) - a(x{-}h/2)(f(x){-}f(x{-}h))$, divided by h^2, is shown (`weighted_second_difference_limit`) to converge to $a(x)f''(x) + a'(x)f'(x) = (af')'(x)$ — the divergence-form operator emerges as a readout of the discrete flux stencil, not assumed. `constant_coefficient_instance` confirms consistency with Step 2.

Step 4 (Tier-0): the off-diagonal dominance criterion, sharp both ways. An anisotropic (cross-term-bearing) quadratic form is representable as a nonnegative-weighted graph form — i.e. as a genuine graph Laplacian — if and only if it is diagonally dominant (`repr_pos_iff_dominant`, `repr_neg_iff_dominant`). Sufficiency is a direct decomposition; necessity is extracted by evaluating any hypothetical representation at three probe vectors, forcing the cross-edge weight to a specific value that goes negative exactly when dominance fails — a genuine obstruction, not an artifact of one decomposition choice.

What this closes and what remains open, by design. Steps 1–4 together give the diagonal-metric Laplace–Beltrami operator class $\sum_i \partial_i(a_i(x)\partial_i u)$ — covering Schwarzschild-radial, FRW, and essentially every metric weak-field tests use — with a sharp criterion for when off-diagonal terms admit a graph representation at all. What remains open, stated at its narrowest: spectral convergence for a fully general (non-dominant) anisotropic metric, and general-metric convergence on a genuine manifold. The latter is a known, substantial result in the literature (Burago–Ivanov–Kurylev, 2014) [15]; it is cited here as prior literature, not mechanized, and is not claimed as this work's contribution.

14 A Discrete Disk-Before-Lock Analog

A discrete, graph-native analog of an unrelated preprint's continuum claim ("Disk Before Mass," a PDE result stated with a provenance map but no worked derivation, hence `Dr` tier and not relied upon here) is built from scratch using only this repository's own retention machinery. Under an explicit *anisotropy hypothesis* — edges classified "perpendicular" or "parallel," with the perpendicular class's benefit always strictly below its strain, and the parallel class's always at or above — a perpendicular candidate edge is shown to strictly worsen the retention functional while a parallel candidate never does, from any current graph state, unconditionally. This is an immediate, per-decision fact, not an asymptotic claim about many growth steps: `InfoDiskBeforeLock.v` (5 theorems, `Th_coqc`). The anisotropy hypothesis itself is disclosed as input data, not derived.

15 Closing the Seed-to-Ceiling Chain

A separate card (Supplement) tracks a chain from the seed's own retention criterion through curvature to an internal spectral ceiling. Two remaining gaps in that chain are closed here. `InfoGrowthFold.v` (5 theorems, `Th_coqc`) generalizes the single-edge curvature-shift law of §10 to an arbitrary *list* of added edges via an exact, order-independent closed form (`forman_fold`), and shows retention only ever bends curvature in one direction regardless of growth order. `InfoCeilingMonotone.v` (6 theorems, `Th_coqc`) shows growth never decreases the quadratic form a Rayleigh quotient is built from, that every exact Rayleigh level achieved before growth is still dominated after growth, and that the degree data feeding the spectral ceiling of §10 only grows — explicitly disclosing what is *not* claimed (individual eigenvalue monotonicity/interlacing, which needs spectral theory not invoked here). Together these make the seed-through- ceiling portion of that chain gap-free up to one disclosed ansatz and one pre-existing, direction-undecidable axiom relating an internal oscillation period to mass — neither of which is discharged by this section, and neither of which is claimed to be.

16 Algebraic Bridges: Curvature Symmetry, Rotation, and a Pentagon Spectrum

Three results connect the curvature/growth track to a separately-developed algebraic track (complex roots of unity, the golden ratio) via concrete shared objects, not analogy. `InfoCurvatureNoether.v` (3 theorems, `Th_coqc`) proves Forman curvature is invariant under any graph automorphism with a correspondingly permuted edge list, extending an existing Noether-style invariance result (proved there only for the Laplacian-built quadratic form) to curvature specifically for the first time; a concrete instance reuses an existing order-6 rotation witness, with the honest caveat that this particular witness graph is regular, so its curvature is trivially constant there — the general theorem is what matters for non-regular graphs. `InfoModeRotation.v` (8 theorems, `Th_coqc`) shows the mother equation's own leapfrog/CFL stepper matrix, at the specific stability-window parameter value where its trace is zero, is *literally* the standard matrix representation of multiplication by i (verified: the same squares-to-negation, period-4 identity already known for i), and that two further window points give periods 6 and 3 — instances of the crystallographic restriction theorem, cited as prior literature for the general claim (only $n \in \{1, 2, 3, 4, 6\}$ are reachable by a rational stepper parameter), not re-derived here. `InfoPentagonSpectrum.v` (5 theorems, `Th_coqc`) proves the 5-cycle graph Laplacian satisfies an exact cubic operator identity whose nonzero-eigenvalue factor has the same defining-relation shape as the golden ratio's own characterizing equation (both are minimal polynomials of elements of $\mathbb{Q}(\sqrt{5})$), and exhibits all four distinct eigenvalues of the 6-cycle (the same graph used as the automorphism witness above) via explicit rational eigenvectors — the 6-cycle is, in this precise sense, the last cycle whose full spectrum stays rational; the 5-cycle is the first whose spectrum leaves $\mathbb{Q}$.

17 A Graph-Native Boundary-Energy Fact and a Degree-Curvature Bound

Two small, self-contained results, explicitly not claimed as more than what they state. `InfoAreaLaw.v` (3 theorems, `Th_coqc`) proves that if a field is constant across a region of a graph, every edge strictly interior to that region contributes exactly zero to the quadratic energy form — all of a uniform region's energy comes from its boundary. This shares a qualitative flavor with the Bekenstein–Hawking holographic bound ("boundary determines the budget, not the interior") but is not that bound and makes no claim about entropy, metric area, or general relativity; sharing a qualitative flavor is not sharing a theorem. `InfoDegreeFromCurvature.v` (2 theorems, `Th_coqc`) proves a node's degree is bounded by 4 minus the local Forman curvature of any incident edge, and, given a curvature floor over a graph, that the maximum degree is bounded by 4 minus that floor — a purely combinatorial fact usable to feed the spectral ceiling of §10 with a curvature-derived value. Stated plainly: given the file's own definition `forman(e) := 4 - deg(u) - deg(v)`, the local bound is an immediate one-line rearrangement (equivalent to non-negativity of degree); the contribution is packaging this as a directly-usable input to the spectral ceiling, not a deep new fact about curvature.

18 General Tensor Reconstruction and a Conditional Curvature Bridge

`InfoTensorFrame.v` (7 theorems, `Th_coqc`, division-free over $\mathbb{Q}$) proves a general, dimension-independent linear-algebra fact: a symmetric bilinear form on an n-dimensional space is fully and uniquely reconstructible from its n axis (diagonal) evaluations plus its $\binom{n}{2}$ face-diagonal evaluations, via the polarization identity — the counting identity $2n + n(n-1) = n(n+1)$ confirms no slack in any dimension, and in 4D, 4 axes + 6 face-diagonals = 10 matches a rank-2 symmetric tensor's component count exactly. This is stated plainly as an *abstract* linear-algebra fact, deliberately not identified with general relativity's metric or stress-energy tensor: no metric signature, no coordinate transformation law, and no field equation is claimed or used here, and the fact holds for a symmetric bilinear form on *any* n-dimensional $\mathbb{Q}$-vector space, gravitational or not. It also does not by itself establish that any graph actually used elsewhere in this work contains face-diagonal edges (the n-D ladder of §13 uses axis-aligned edges only).

`InfoStrainTensorBridge.v` (11 theorems, `Th_coqc`) connects this to curvature on a stated condition: on a 2×2 graph cell with a diagonal candidate edge, the strain on that edge equals the tensor-frame diagonal evaluation *exactly when the cell is affine* (no discrete Hessian cross term), and differs from it by an exact, fully disclosed defect term in general. Under that affine condition, the seed's own retention criterion for the diagonal candidate is shown to be, verbatim, a price on a tensor-frame evaluation (`diagonal_retention_prices_tensor`) — curvature and this tensor-reconstruction fact are connected by a real theorem for the affine case, not merely both being expressible on the same graph. This is *not* an unconditional closure; for a non-affine cell the disclosed defect term must be carried. The same file separately proves that any graph's quadratic form splits exactly into inside, outside, and cut contributions against any region (`gform_screen_partition`, unconditional), and, under one further disclosed lens (benefit per retained distinction taken proportional to a temperature-like parameter), that the retention criterion becomes a discrete Clausius-type inequality, both directions exact. None of this constitutes or is claimed to constitute Jacobson's thermodynamic derivation of the Einstein equation [18]; it is a small, honestly-tiered fragment sharing a qualitative shape with one step of that program, explicitly not the program itself.

19 Momentum Optimizer Theory Coincides With the Kernel's Energy Theory on a Quadratic Mode

`InfoOptimizerWindow.v` (10 theorems, `Th_coqc`, division-free over $\mathbb{Q}$) shows the classical heavy-ball/momentum iteration on a quadratic mode is, under an exact reparametrization, the same dissipative two-step recurrence already proved for the mother equation's own leapfrog stepper in §10. The reparametrized Lyapunov functional decrements by exactly one nonnegative square per step; a momentum dichotomy (never increases below a threshold, exactly conserved at the threshold, never decreases above it) is proved as a theorem; and an a-priori bound on every iterate is obtained from initial data alone, with no spectral computation or run-time tuning, inside the classical stability window. Divergence outside that window, convergence rates, and anything about non-quadratic objectives are explicitly not claimed.

20 A Composed Reading: Curvature-Limited Mass

A companion synthesis note (`paper/mass_note.tex` in this repository, also independently authored by the present author, cross-checked against this repository at the exact commit this manuscript itself was built from) composes several of the theorems above — none of them individually about mass — into a mass chain: retention → degree → curvature → an internal-mode frequency ceiling → mass, through exactly one disclosed, non-theorem identification. We state the composition here at main-text weight because it is a genuine, checkable consequence of results already established above, not a new file; the companion note carries the full derivation, literature positioning, and a 21-file SHA-256-pinned artifact table.

The Anderson–Morley eigenvalue ceiling (Anderson & Morley, 1985) is now mechanized directly, in witness form over $\mathbb{Q}$ (`InfoSpectralCeilingSharp.v`, 2 theorems plus 4 supporting lemmas, all `Th_coqc`; `anderson_morley_witness`): for any exact eigenpair, some edge (u, v) satisfies $|\lambda| \leq \deg_E(u) + \deg_E(v)$, proved by taking the edge maximizing $|x_u - x_v|$ and comparing the eigen-equation at its two endpoints directly — no edge-space Gershgorin matrix, no division. Composed with the exact identity $\deg_E(u)+\deg_E(v) = 4-F(e)$ (`InfoDegreeFromCurvature.v`) under a curvature floor, this gives the *sharp* curvature ceiling directly (`sharp_curvature_ceiling`), $\lambda \leq 4 - F_{\min}$ — a factor $\sqrt{2}$ tighter, at the frequency level, than the coarser $2d_{\max}$ route this manuscript mechanized earlier (§10, `InfoSpectralCeiling.v`), which remains correct but is no longer the sharpest available bound. The mass ceiling below now uses the sharp form throughout:

$$\omega_{\rm int}^2 \;\leq\; \frac{K}{M}\,(4 - F_{\min}), \tag{17}$$

a direct algebraic substitution of two already-`Th_coqc` facts, itself `Th_coqc` (no additional Coq file needed: both inputs are theorems, and the composition is one inequality chain). Given exactly one imported, disclosed identification — $\tau_c := 1/\omega_{\rm int}$ together with the programme's own mass–clock postulate $m = \hbar/(2\tau_c c^2)$ (a programme postulate, not a published result, tagged `[AX/import]` in the companion note, selected over two explicitly rejected sign-analysis alternatives there) — Eq. 17 becomes a curvature-limited mass bound,

$$m \;\leq\; \frac{\hbar}{2c^2}\sqrt{\frac{K}{M}\,(4 - F_{\min})}. \tag{18}$$

This bound is `Th_coqc` *conditional on* the one imported identification, and holds as stated *for any excitation that is an exact mode of the graph carrying it*: the inequality it rearranges is fully mechanized, but Eq. 18 itself is not a standalone Coq theorem (no `Print Assumptions` block asserts it directly) — it is the companion note's own arithmetic on top of (17), disclosed as such there. A second, independent caveat applies to the physically motivating case of an *approximately localized* excitation restricted to a support subgraph rather than an exact mode of the whole graph: the companion note states plainly that this restriction is, at present, only a numerical finding (a diagnostic mode carrying 99% of its weight on a dense cluster across an ensemble of random graphs), `finite_diagnostic`, not a kernel theorem even granting the ansatz above — and we repeat that distinction here rather than let the theorem-conditional-on-ansatz framing imply more than it does for the localized case, which is exactly the case the "curvature license" reading below is really about. Its contrapositive is the note's sharper claim: mass exceeding the $F_{\min} = 0$ cap $(\hbar/2c^2)\sqrt{4K/M}$ forces negative curvature somewhere on the excitation's support — curvature is not sourced by mass here, it is a capacity mass requires, the reverse of general relativity's habitual direction of explanation, and inherits both caveats above. Both the bound and its contrapositive additionally inherit a disclosed dependence on whether Forman curvature deserves the name at all — the same open question already carried by this manuscript's Open Problem 3 (§27) and named OB-FORMAN-RICCI in the companion note — neither claim is stronger than that dependence allows, and the companion note states this explicitly rather than eliding it.

21 An Entropy-Language Restatement, and a Boundary-Screening Theorem

Two further small, self-contained results, each explicitly scoped to state no more than what it proves.

`InfoEntropyLicense.v` (6 theorems, `Th_coqc`, division-free over $\mathbb{Q}$) restates the degree/frequency-ceiling family of §10 in entropy-flavored language: defining a per-node quantity $S_{\rm loc}(s_0, E, i) := s_0 \cdot \deg_E(i)$ for an arbitrary weight s_0, the handshake identity becomes $\sum_i S_{\rm loc}(s_0, E, i) = 2s_0|E|$; $S_{\rm loc}$ is proved monotone non-decreasing under retention-only graph growth (the same $E_s{+}{+}E$ shape as §11); and, given the "achieving" hypothesis $2d_{\max} = \deg_E(u) + \deg_E(v)$ (an explicit hypothesis, of exactly the same kind §10's own spectral ceiling already takes for $d_{\max}$), the frequency-ceiling inequality $M\omega^2 \leq K \cdot 2d_{\max}$ composes *genuinely* with the entropy identity into a bound on the entropy-language quantity itself, $M\omega^2 \leq K \cdot (S_{\rm loc}(s_0, E, u) + S_{\rm loc}(s_0, E, v))/s_0$ — not a mere side-by-side pairing of two unrelated facts, but one inequality substituted into the other. A further rearrangement gives a lower bound on $S_{\rm loc}$ at the degree-realizing node in terms of a plain-$\mathbb{Q}$ energy relation. We state plainly what this is and is not: s_0 is an arbitrary $\mathbb{Q}$-parameter and $S_{\rm loc}$ is, by definition, a rescaling of degree; calling it "entropy" is a naming choice in the same family as the qualitative Bekenstein–Hawking resemblance already disclaimed for the boundary-energy fact of §17, not a new physical claim, not the Bekenstein bound [19], and not an instance of Wheeler's "it from bit" [20]. No thermodynamic, statistical-mechanical, or information-theoretic (in the Shannon or Szilárd–Landauer bit-erasure [21, 22] sense) identification of s_0 or $S_{\rm loc}$ is claimed or used anywhere in the file.

`InfoBoundaryScreening.v` (5 theorems, `Th_coqc`) proves a graph-native screening fact on top of `InfoStrainTensorBridge.v`'s region-partition identity of §18 (`gform_screen_partition`): calling the outside-plus-cut part of the quadratic form against a region *Exterior*, and calling a node *strictly interior* when it is in the region and every incident edge of the graph stays inside the region, Exterior's value is exactly unchanged by any change to a field's values confined to non-interior nodes—two fields that agree everywhere off the strict interior give *identical* Exterior values, unconditionally and exactly, for any graph, region, and pair of fields (`exterior_invisible_to_ interior_swap`). The proof is a direct case analysis on which of the region-partition's three edge classes (inside/outside/cut) each edge falls into: every edge contributing to Exterior has at least one endpoint outside the region, and that very edge already witnesses that both its endpoints fail the strict-interior test, so the fixed weighting is untouched when the field is redefined there. We state plainly what this does and does not establish. It *does* show that, on this graph's own native quadratic form, information about the field's values in the strict interior cannot propagate into Exterior — a purely combinatorial, unconditional fact about which edges carry a given field to the outside. It does *not* establish anything about physical entropy, the holographic principle [23, 24], a Bekenstein-Hawking area law [19], or general relativity; "boundary", "screening", "Exterior", and "capacity" (the trivially field-independent cut-edge count, stated for completeness with a one-line proof) are graph-native names chosen for readability, not physical identifications, in the same spirit as §17's explicit disclaimer that a qualitative resemblance to the Bekenstein–Hawking bound is not that bound.

22 Opening the Nonlinear Sector, and a Readability-Boundary Instance

Two further small results, one opening genuinely new ground for this kernel, one closing one more instance of an existing algebraic-restriction story.

`InfoCubicLinearization.v` (6 theorems, `Th_coqc`, division-free over $\mathbb{Q}$) is the first file in this arc to give the mother equation's $\nabla V(\Phi)$ slot a concrete, nonzero, mechanized instance: every prior result in this manuscript works in the conservative, purely quadratic sector ($\nabla V = 0$). Adding a single on-site cubic term $-g\Phi^3$ (from $V(\Phi) = (g/4)\Phi^4$) to the existing linear force, the file proves the exact one-step polarization identity for the force difference under a field perturbation, extracts the linear-order coefficient shift $q(i) := 3g\,\Phi(i)^2$ that the perturbation acquires from the background, and its sign ($q \geq 0$ for $g \geq 0$, strictly positive on the background's support for $g > 0$). What is *not* claimed, stated in the file's own header: any time-averaging or homogenization step (the passage from a pointwise coefficient shift to an effective-medium mass reading), and any propagation or spectral consequence — those remain explicitly disclosed, not mechanized. A companion numerical probe (not part of this manuscript's own claim set, reported in the project's supplementary material) found that locking energy into a zero-group-velocity mode of a region measurably slows a passing low-frequency packet only when $g \neq 0$ (a linear, $g = 0$ control shows no effect, consistent with superposition); this is offered as motivation for why the ∇V term matters, not as something this file itself establishes.

`InfoReadabilityBoundary.v` (3 theorems, `Th_coqc`, plus one worked example and one small composing remark) proves that no rational parameter of §16's stepper matrix makes its 5th iterate the identity: the relevant polynomial factors exactly as $-(a^2-5a+5)(a^2-3a+1)$ (an exact ring identity), and both quadratic factors have discriminant 5, so a rational root would force a rational square root of 5 — proved impossible from scratch via integer descent. This is one further instance in the same family as §16's periods 3, 4, and 6, but is stated narrowly: it does *not* establish that $\{1, 2, 3, 4, 6\}$ is the complete set of rational periods reachable by this stepper, which would require the general crystallographic restriction theorem (cited as literature, not mechanized here) — an earlier, broader draft of this result was reviewed, found not to follow from the rest, and dropped rather than weakened into an inaccurate statement.

23 Reviewer Attack Response

This section pre-empts questions an adversarial referee will ask, following an independent technical review of an earlier draft.

Is $E^2M = \hbar^2K\lambda$ really quantum mechanics? No, not non-relativistic Schrödinger mechanics (correctly flagged: that theory is first-order in E). It is a relativistic (Klein–Gordon-family) dispersion, and §6 proves this is exact, not a relabeling.

Is the Hückel match a real validation? It is a consistency demonstration of graph-topology sensitivity (confirmed via a non-aromatic control), not an independent confirmation of a fitted physical constant (§4).

Is Eq. 6 really "frame covariance"? It is invariance under edge-list relabeling, a bookkeeping symmetry, not physical boost covariance; the latter (Eq. 7's invariance under the imported boost formula) is stated and proved separately and is not conflated with it here.

Is sgn *in Eq. 4 actually derived from the causal order $\prec$?* No — it is a free parameter in the source theorems. This manuscript states that explicitly (§5) rather than implying otherwise, and lists deriving it as open work (§27).

Is the curvature–dissipation link (Eq. 12) non-trivial? It is an exact algebraic identity, but shallow: it holds because both quantities are stated in terms of degree, not because curvature causes a dissipation requirement. This is disclosed, not obscured.

Are all citations independently checkable? Every citation in §28 was checked against standard physics/mathematics references; two citations flagged as unresolved in an internal draft (an anonymous arXiv preprint and a mismatched preprint server) have been removed from this version pending independent verification, rather than retained on trust.

Does the frequency ceiling (Eq. 15) claim a physical Planck-scale minimal length? No. It is a structural/discretization bound: a property of L_R's finite maximum degree. It states *that* a ceiling exists and *exactly* what sets it ($d_{\max}$, M, K), not a numerical value for any particular physical system.

Is the causal-signature closure (§10) actually complete? No, and the manuscript says so directly: the construction is proved for an arbitrary relation with genuine incomparable pairs, demonstrated on a constructed example, but this repository's own causal order is a total order and cannot host one. This is reported as a partial result with a named obstruction, not a closed gap.

Does the graph-growth result (§11) address the Hubble tension or any cosmological discrepancy? No, explicitly. That is a general- relativistic problem and §7 establishes no general relativity is derived here. The growth result is offered only as a native discrete analog of expansion, with an honestly bounded scope (power-law dilution only; no de Sitter behavior found; topology-dependent, not universal).

Does the boundary-energy fact (§17) establish a holographic entropy bound? No. It shares a qualitative flavor with the Bekenstein–Hawking bound (boundary determines the budget, not the interior) but proves a purely combinatorial fact about a quadratic energy form on a graph — no entropy, no metric area, and no general-relativistic structure appears in the statement or the proof.

Does §18's tensor-reconstruction fact establish anything about physical tensors (metric, stress-energy), and is the curvature bridge to it unconditional? No on both counts. The reconstruction fact is an abstract linear-algebra identity holding for any symmetric bilinear form on any n-dimensional $\mathbb{Q}$-vector space; it is not applied to, or claimed to be, general relativity's metric or stress-energy tensor. The curvature bridge (`diagonal_retention_prices_tensor`) holds only on the stated affine-cell hypothesis; for a non-affine cell an explicitly disclosed defect term must be carried, and this is not an unconditional closure of any kind.

Is the curvature-limited mass bound (§20) itself a Coq theorem? No, and this manuscript says so directly. The inequality it algebraically rearranges (Eq. 17) is a direct composition of two already-mechanized theorems and is itself `Th_coqc`; the mass bound (Eq. 18) additionally depends on one imported, non-theorem identification ($\tau_c := 1/\omega_{\text{int}}$ plus the imported mass–clock relation), so it is conditional, not unconditional `Th_coqc` — exactly as labeled. No new Coq file mechanizes Eq. 18 as a single theorem. A second, lower-tier caveat applies specifically to localized (particle-like) excitations restricted to a support subgraph rather than an exact whole-graph mode: that restriction is currently only a numerical (`finite_diagnostic`) finding, not a theorem, even granting the ansatz above — stated in §20 and repeated here because it is precisely the case the mass narrative is about.

24 Novelty Audit

The individual mathematics throughout — Gershgorin/Rayleigh bounds, leapfrog invariants, discrete Green's identities, Forman curvature, lattice Noether theorems, variational integrators, Kronecker sums, diagonal-dominance criteria — is classical or already in the literature, and this repository's own discrete-substrate-physics framing sits in a crowded field (causal sets, the Wolfram Physics Project, Regge calculus, loop quantum gravity). A dedicated literature check (2026-07-05) on the five specific structural identifications introduced in §12–11 found no close prior art for three of them (the curvature-affine horizon threshold; the horizon dichotomy proved rather than stipulated; an epistemic-axiom-to-energy-floor bound, not mechanized here), one close but distinct prior construction requiring explicit citation (the double matter/geometry stationarity, §12, cf. Trugenberger's Combinatorial Quantum Gravity [16]), and one instance of well-established qualitative folklore given a specific proved form here for the first time as far as this check found (the order-collapse theorem, §11, cf. causal set theory [17]).

Scope limit, stated plainly. This literature check covers only §12–11. The eleven files and six sections added later (§14–19) have NOT been through an equivalent dedicated check and their novelty is therefore *unaudited*: several of their individual ingredients are flagged in-line as classical or adjacent to known results (the crystallographic restriction theorem cited as prior literature in §16; the Bekenstein–Hawking qualitative resemblance disclaimed in §17; the polarization identity underlying §18), but no systematic search for close prior art was performed for these sections as a whole. §20's composed mass reading is a partial exception: the companion synthesis note that develops it independently positions each constituent equation against ten literature streams by owner and states its own search protocol for the narrower claim that no prior discrete-gravity programme possesses a machine-checked kernel of any size; that audit is the companion note's own and is not reperformed or independently re-verified here.

The claimed contribution is therefore narrower than "new physics" and is stated at that level: a single graph operator carrying genuine (not aliased) derivations — not results sharing a vocabulary, but a shared equation — across dispersion, causal structure, conservation, symmetry, curvature, and variational structure simultaneously, machine-checked and axiom-free end to end, with a tier system that separates theorem from stance at every step. This is a claim about *method and integration*, not about any single physical prediction; the one structural prediction native to this framework is the frequency/UV ceiling of §10 (Eq. 15), which has not been tested against any experiment.

25 Main-Text vs. Supplement Boundary

This manuscript is self-contained for its theorem-level claims: every stated equation carries its exact source file and tier, and §26 gives the exact commands to independently re-check every one. The manuscript is responsible for the mother equation, the four branch results (§4–8), the unification result (§6), the quasinormal-mode bridge (§9), the strengthening campaign (§10), graph growth (§11), the action-stationarity and curvature-horizon structure (§12), the n-D spectral ladder (§13), the disk-before-lock analog (§14), the seed-to-ceiling closure (§15), the algebraic bridges (§16), the boundary-energy fact and degree-curvature bound (§17), the tensor-reconstruction and conditional curvature bridge (§18), the momentum-optimizer identification (§19), the composed curvature-limited mass reading (§20), the entropy-language restatement and boundary-screening theorem (§21), the reviewer attack response (§23), and the reproduction commands.

The companion synthesis note (`paper/mass_note.tex`, compiling to `paper/mass_note.pdf`) is a separate, self-contained preprint by the same author, included here as a supplement because it is built directly on this repository's own theorems at this manuscript's own commit; it carries its own abstract, literature review (ten streams, positioned individually), circularity audit, and open-problem register, and should be read as a companion document, not as part of this manuscript's own claim set. Where the two overlap (§20 above), this manuscript's statement is the authoritative one for this repository's own claim set; it is not, on every point, strictly narrower than the companion note's — §20 now carries both of the companion note's caveats on the mass bound (the ansatz dependence and the separate localization-to-support-subgraph caveat) precisely so this manuscript is not less conservative than the note it summarizes.

The companion supplement (`SUPPLEMENT.md`) is responsible for material this manuscript summarizes but does not reproduce in full: the complete dependency DAG in both diagram and prose form; the full log of all eight attempts to derive general relativity, including the ones only briefly mentioned in §7; the complete novelty audit against prior discrete-substrate literature (§24 gives only its conclusion); and the per-citation verification notes recording how each reference in §28 was checked, including two citations present in an earlier internal draft and removed after they could not be independently verified (§23). A reader who wants the theorem statements and can reproduce them needs only this manuscript; a reader who wants the full argument trail, including dead ends, needs the supplement as well.

26 Reproducibility Protocol

1. `git clone` this repository.
2. `make verify` — compiles all 37 `formal/*.v` files via `coqc` in dependency order and runs `Print Assumptions` on every named theorem (199 total), printing a tier report (`Th_coqc` / `+reals` per file).
3. `python3 scripts/verify_quantum_gravity_root_bridge.py` — reproduces the quasinormal-mode convergence table (Eq. 14).
4. Expected result: all Coq files compile with exit code 0; the tier report matches Table 1; the QNM script's $N = 3200$ row matches Eq. 14 to the printed precision ($N = 1600$ is already close, $0.4838 - 0.0958i$, but

the digits quoted in Eq. 14 are the $N = 3200$ row).

27 Open Problems

(1) Construct a genuinely non-total (partial, multi-dimensional) causal order on this repository's graphs, so that the sign-construction theorem of §10 produces a non-degenerate indefinite signature on the repository's *own* causal structure, not only on a constructed example. (2) Extend the quasinormal-mode bridge (§9) to spin-2 perturbations. (3) Determine whether Ollivier–Ricci curvature [7] offers a non-shallow alternative to Eq. 12's exact-but-shallow link. (4) Does the dilution law of §11 admit a principled selection rule for which growth topologies are physically relevant, or is topology-dependence the final answer? (5) Does §10's Noether pairing admit a principled dissipative correction, or is exact conservation strictly a conservative-sector fact? (6) Discharge, or find a counterexample to, the pre-existing, direction-undecidable axiom relating an internal oscillation period to mass that §15 leaves untouched — the seed-to-ceiling chain is gap-free only up to this axiom. (7) Derive, rather than disclose as input data, the anisotropy hypothesis of §14 (which edges count as "perpendicular" vs. "parallel" and why) from a more primitive graph property, so the disk-before-lock result no longer depends on an externally-supplied classification. (8) Prove or refute an inertia cost law (cost of re-addressing the internal loop of §20 scales as $\propto m$ under acceleration); the companion synthesis note states this is currently unproven, not merely uncited, and flags it as its own furthest interpretive claim.

28 Conclusion

One graph-Laplacian root equation genuinely derives, and genuinely unifies, a quantum dispersion relation and a special-relativistic wave operator. It honestly does not derive general relativity, for a stated philosophical reason rather than a numerical shortfall, and offers in its place a native discrete curvature readout with an exact, if shallow, link to an independently-proven stability result. Responding to adversarial review, six further theorems promote a frequency ceiling, an energy-balance law, a first-order unitary-like skeleton, a causal sign construction (partially closed, honestly), and a discrete Noether law from informal stances to machine-checked facts; a seventh shows that graph growth is a native, non-continuum, honestly-bounded analog of expansion, explicitly not a claim about any physical cosmological discrepancy. Two further results prove the mother equation is an exact stationarity readout on both its field and its geometry, and compose the graph's own curvature into an exact horizon threshold and dichotomy — reported at full strength for what they establish, and explicitly not as evidence of narrowing any gap toward general relativity's field equations, since variational-structure-sharing is generic to essentially all physical theories. A final four-step ladder closes the flat and diagonal-metric cases of this framework's n-dimensional continuum limit, tagging each step at its true tier and narrowing the remaining general-metric case to a single, named, literature-cited gap. Eleven further files extend the reach without ever advancing a claim past its evidence: a discrete anisotropy result immediate at every decision point, not asymptotic; two files closing a seed-to-spectral-ceiling chain up to one disclosed ansatz and one pre-existing, direction-undecidable axiom, neither discharged here; three files bridging curvature and growth to a separately-built algebraic track via concrete shared objects (an automorphism-invariance extension, a literal identity between the mother equation's own stepper matrix and the standard i-rotation matrix, and a pentagon-graph spectrum landing in the same field as the golden ratio); a graph-native boundary-energy fact and a degree-curvature bound, each explicitly scoped against the physical results (holographic entropy, general relativity) they share only a qualitative flavor with, not a theorem; a general tensor- reconstruction fact and a curvature bridge to it that is conditional on one named hypothesis, not unconditional; and a momentum-optimizer identification inheriting the mother equation's own energy theory by exact reparametrization. Every claim above carries its tier and its exact reproduction command.

Acknowledgements

This work was developed through a human–AI co-thinking workflow, including one round of independent adversarial technical review external to the author's own assessment. Any remaining errors are the author's own.

Author Declarations

Funding. No specific grant from any funding agency in the public, commercial, or not-for-profit sectors was received for this research.
Competing Interests. The author declares no competing interests.
AI Assistance Disclosure. AI tools (Claude, Anthropic) were used for formal-proof drafting, literature cross-checking, numerical implementation, and editorial structuring, including an independent adversarial review pass. Substantive claims, tier assignments, and final decisions remain the author's responsibility.
Data and Code Availability. All Coq sources,

the Python reproduction script, and this manuscript's source are available in the accompanying repository.

References

[1] M. Planck, Über das Gesetz der Energieverteilung im Normalspectrum, *Annalen der Physik*, 1900.

[2] A. Einstein, Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt, *Annalen der Physik*, 1905.

[3] A. Einstein, Zur Elektrodynamik bewegter Körper, *Annalen der Physik*, 1905.

[4] H. A. Lorentz, Electromagnetic phenomena in a system moving with any velocity smaller than that of light, *Proc. Acad. Science Amsterdam*, 1904.

[5] E. Hückel, Quantentheoretische Beiträge zum Benzolproblem, *Zeitschrift für Physik*, 1931.

[6] R. Forman, Bochner's method for cell complexes and combinatorial Ricci curvature, *Discrete & Computational Geometry*, 2003.

[7] Y. Ollivier, Ricci curvature of Markov chains on metric spaces, *Journal of Functional Analysis*, 2009.

[8] T. Regge, J. A. Wheeler, Stability of a Schwarzschild singularity, *Phys. Rev.*, 1957.

[9] E. Leaver, An analytic representation for the quasinormal modes of Kerr black holes, *Proc. R. Soc. Lond. A*, 1985.

[10] E. Berti, V. Cardoso, A. O. Starinets, Quasinormal modes of black holes and black branes, *Class. Quantum Grav.*, 2009.

[11] J.-P. Bérenger, A perfectly matched layer for the absorption of electromagnetic waves, *J. Comput. Phys.*, 1994.

[12] L. Bombelli, J. Lee, D. Meyer, R. Sorkin, Space-time as a causal set, *Phys. Rev. Lett.*, 1987.

[13] S. Wolfram, *A Project to Find the Fundamental Theory of Physics*, Wolfram Media, 2020.

[14] T. Regge, General relativity without coordinates, *Il Nuovo Cimento*, 1961.

[15] D. Burago, S. Ivanov, Y. Kurylev, A graph discretization of the Laplace-Beltrami operator, *J. Spectral Theory*, 2014.

[16] C. A. Trugenberger, Combinatorial quantum gravity: geometry from random bits, *JHEP* 09:045, 2017.

[17] S. Surya, The causal set approach to quantum gravity, *Living Reviews in Relativity*, 2019.

[18] T. Jacobson, Thermodynamics of spacetime: the Einstein equation of state, *Phys. Rev. Lett.* 75, 1260, 1995.

[19] J. D. Bekenstein, Black holes and entropy, *Phys. Rev. D* 7, 2333, 1973.

[20] J. A. Wheeler, Information, physics, quantum: the search for links, in *Complexity, Entropy and the Physics of Information*, Addison-Wesley, 1990.

[21] L. Szilárd, Über die Entropieverminderung in einem thermodynamischen System bei Eingriffen intelligenter Wesen, *Zeitschrift für Physik* 53, 840, 1929.

[22] R. Landauer, Irreversibility and heat generation in the computing process, *IBM J. Res. Dev.* 5, 183, 1961.

[23] G. 't Hooft, Dimensional reduction in quantum gravity, in *Salamfestschrift*, World Scientific, 1993; arXiv:gr-qc/9310026.

[24] L. Susskind, The world as a hologram, *J. Math. Phys.* 36, 6377, 1995.